



PRODUCT NAME

Industrial-grade Hair Conditioning Cream Formula

Description

This industrial-grade hair conditioning cream is expertly formulated to provide exceptional moisturizing and detangling benefits, while maintaining a perfectly balanced pH level of 7. It is designed to deliver high viscosity for effortless application and optimal hair care performance.

Technical Specifications

pH Level:	7
Viscosity:	High
Shelf Life:	24 months
Batch Size:	50-100 kg
Processing Time:	2 hours
Temperature:	75°C
Storage Conditions:	Store in a cool, dry place away from direct sunlight.
Equipment:	Mixing vessel, Heating equipment, Homogenizer
Certification:	ISO 22716:2007 (GMP for cosmetics)

Formulation Table

Sr.No	Ingredient	INCI Name	%	Function
1	Water	Aqua	65.0%	Solvent
2	Cetearyl Alcohol	Cetearyl Alcohol	10.0%	Emollient, Thickener
3	Behentrimonium Chloride	Behentrimonium Chloride	8.0%	Conditioning Agent, Detangling
4	Glycerin	Glycerin	5.0%	Humectant
5	Cyclopentasiloxane	Cyclopentasiloxane	5.0%	Emollient, Detangling
6	Phenoxyethanol and Ethylhexylglycerin	Phenoxyethanol and Ethylhexylglycerin	1.0%	Preservative
7	Citric Acid	Citric Acid	0.5%	pH adjuster
8	Fragrance	Parfum	0.5%	Fragrance
9	Dimethicone	Dimethicone	5.0%	Emollient
Total			100%	

Manufacturing Process

Phase 1: Aqueous Phase

1. Heat water (Aqua) to 75°C in a main vessel.
2. Add Glycerin and Citric Acid to the heated water and stir until dissolved.

Phase 2: Oil Phase

1. In a separate vessel, combine Cetearyl Alcohol, Behentrimonium Chloride, Cyclopentasiloxane, and Dimethicone.
2. Heat to 75°C and mix until homogenous.

Phase 3: Emulsification

1. Slowly add the Oil Phase to the Aqueous Phase while continuously stirring.
2. Maintain temperature and mix for an additional 15 minutes to ensure a stable emulsion.

Phase 4: Cooling Phase

1. Begin cooling the mixture while stirring slowly.
2. At 40°C, add Phenoxyethanol and Ethylhexylglycerin, and Fragrance.
3. Continue to stir until the mixture reaches room temperature.

Required Equipment

Mixing vessel, Heating equipment, Homogenizer

Safety Precautions

Handling: Wear appropriate PPE including gloves, safety glasses, lab coat.

PPE Requirements: Chemical-resistant gloves, safety goggles, protective clothing.

Storage: Store in cool, dry place away from direct sunlight. Keep containers tightly closed.

Storage Conditions: Store in a cool, dry place away from direct sunlight.

Packaging Notes

Packaging: Use chemically compatible containers (HDPE, glass, or PET).

Labeling: Include product name, ingredients, usage instructions, and safety warnings.

Certification: ISO 22716:2007 (GMP for cosmetics)

Scaling Note

Lab Scale: This formulation is designed for laboratory testing and development.

Pilot Scale: For pilot production, scale proportionally and verify all parameters.

Production Scale: Consider equipment limitations, mixing efficiency, and process validation.

Batch Size: Current formulation is optimized for 50-100 kg.

Scaling Factor: Maintain ingredient ratios while adjusting processing parameters as needed.

Product Usage Instructions & Application Guidelines

Apply a generous amount of the conditioning cream to clean, damp hair. Distribute evenly from roots to tips. Leave on for 3-5 minutes, then rinse thoroughly with lukewarm water.